TT-P-96D
AMENDMENT - 2
June 27, 1983
SUPERSEDING
Amendment - 1
June 23, 1977

FEDERAL SPECIFICATION

PAINT, LATEX BASE, FOR EXTERIOR SURFACES
(WHITE AND TINTS)

This amendment, which forms a part of Federal Specifications TT-P-96D, dated June 11, 1976, was approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

PAGE 1

Under Federal Specifications, add:

TT-E-545 - Enamel, Odorless, Alkyd, Interior-Undercoat, Tints and White

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Under American Society for Testing and Materials (ASTM) Standards, delete:

D 476 - Test for Consistency of Paints Using the Stormer Viscometer
D 3335 - Test for Low Concentration of Lead in Paint by Atomic Absorption Spectroscopy

Add:

D 476 - Specifications for Titanium Dioxide Pigments
D 523 - Specular Gloss
D 1729 - Visual Evaluation of Color Differences of Opaque Materials
D 1734 - Making and Preparing Concrete and Masonry Panels for Testing Finishes
D 2805 - Test Method for Hiding Power of Paints
G 23 - Operating Light - and Water - Exposure Apparatus (Carbon-Arc Type) for Exposure of Nonmetallic Materials

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TABLE I, last line, change to: "Lead, percent by weight of non-volatile 0.06 max for all types"

FSC 8010

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TABLE IV, delete in its entirety and replace with:

TABLE IV. Index

Characteristics	Requirement Reference	Test Method: Fed. Test Method Std. No. 141	Test Method: ASTM Method	Paragraph Reference
Color	3.2.3	----	D 1729	4.4.3
Total solids	Table I	----	D 2369	----
Pigment, percent	Table I	4021	----	4.4.2
Non-volatile vehicle	Table I	4053	----	----
Weight per gallon	Table I	----	D 1475	----
Consistency	Table I	----	D 562	----
Fineness of Grind	Table I	----	D 1210	----
Drying time	Table I	4061	----	----
Reflectance	Table I	----	E 97	----
Gloss, 60° specular	Table I	----	D 523	4.4.16
Dry opacity	Table I, II 3.3.1	----	D 2805	----
Lead percent	Table I	----	----	4.4.17
Condition in container	3.4.1	3011	----	4.4.4
Storage stability	3.4.2	----	----	4.4.5
Application properties	3.4.3	----	D 358 D 1734	4.4.6
Odor	3.4.4	----	D 1296	----
Appearance	3.4.5	----	----	4.4.7
Flexibility	3.4.6	----	----	4.4.8
Freeze-thaw resistance	3.4.7	----	D 562	4.4.9
Biological growth	3.4.8	6271	----	4.4.10
Accelerated weathering	3.4.9	----	G 23	4.4.11
Heat stability	3.4.10	----	----	4.4.12
Recoating	3.4.11	----	----	4.4.13
Water resistance	3.4.12	----	----	4.4.14
Washability	3.4.13	6141	----	4.4.15

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Para. 4.4.3, change: "...method 4250 of Fed. Test Method Std. No. 141..." to "...ASTM D 1729 ..."

Para. 4.4.6.1(3), change: "...method 2051 of Fed. Test Method Std. No. 141..." to "...ASTM D 1734.."

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Para. 4.4.11, lines 5 and 6. change: "...method 6151 of Fed. Test Method Std. No. 141..." to "...ASTM G 23 using type E apparatus with an exposure cycle of 102 minutes light followed by 18 minutes light and water spray."

Para. 4.4.15. Change the paragraph to read:

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4.4.15. Washability (wet abrasion resistance). Test the paint in accordance with method 6141 of Fed. Test Method Std. No. 141, except the paint shall be applied over a 38 - 64 mm (0.0015-0.0025 in) thick dried film of enamel undercoater complying with TT-E-545. The spong shall be recharged every 25 cycles until 750 cycles have been completed. Evaluate for compliance with 3.4.13.

Add new paragraph:

4.4.16. Specular gloss. Draw down the thoroughly mixed paint on plane opaque white glass panels as specified in 2.1.5 of method 2021 of Fed. Test Method Std. No. 141. Use a doctor blade which will produce a wet film thickness of 0.003 ± 0.0001 inch. Dry the panels for 48 hours at standard conditions in a dust free environment. Determine 60° specular gloss in accordance with ASTM D 523.

Add new paragraph:

4.4.17. Lead content.

4.4.17.1 Sample preparation. Using a 0.006-inch film applicator and a mechanical applicator plate, duplicate drawdowns for each sample of well-mixed paint shall be made on a standard paint penetration chart and dried for 24 hours. The drawdown shall be cut into disc of appropriate size to fit the sample holder of a fluorescence X-ray spectrometer.

4.4.17.2 Procedure. Lead content shall be determined using an X-ray flourescence spectrometer capable of determining lead at a minimum level of 0.03 percent by weight of the total nonvolatile. The settings for a wavelength, dispersive fluorescence spectrometer shall be as follows: (1)

Element	Analytical Line	Angle	Crystal	Detection	Collimator	X-ray tube (MO)
Pb	L	33.93	LiF(200)	Flow S.C.	Fine	60kV 45mA
Pb (Backgrd I)		33.00	LiF(200)	Flow S.C.	Fine	60kV 45mA
Pb (Backgrd II)		35.50	LiF(200)	Flow S.C.	Fine	60kV 45mA
Mo	K	20.33	LiF(200)	Flow S.C.	Fine	60kV 45mA

Pulse height selection shall be used in all measurements and counting time shall be 100 seconds. Place the sample disc in the wavelength dispersive unit. Measure the count rates of lead, lead background and the Molybdenum Compton scattered background from the X-ray tube.

4.4.17.3 Calculation.

$$\frac{2I_{Pb} - I_{Pb}\ (\text{background I}) - I_{Pb}\ (\text{background II})}{2}$$

where I equals gross intensity. These results shall be compared with those obtained by a 0.06 percent lead standard made up from the same type of paint sample and evaluated for compliance with Table I.

(1) Energy dispersive fluorescence spectrometers shall be set up according to the manufacturer's manual.

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